Practical-8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Theory

1. Is the schema normalized properly?

Answer: Yes. The database is divided into separate tables like students, events, and registrations. This avoids repeating the same data and keeps the database organized.

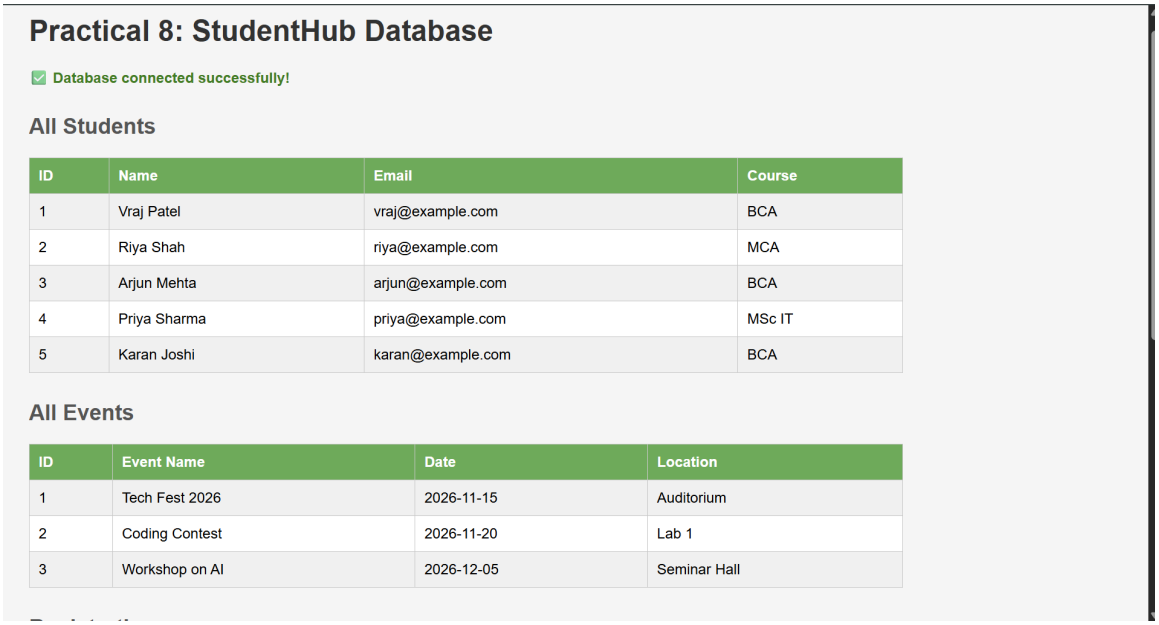
2. Are primary keys and foreign keys correctly defined?

Answer: Yes. Each table has a **primary key** to uniquely identify each record. Foreign keys connect related tables, such as student_id and event_id in the registrations table.

3. Is the database connection handled securely?

Answer: Yes, reasonably. The connection details are kept in a separate db.php file, and the code uses prepared statements when inserting data.

Result:



Practical 8: StudentHub Database

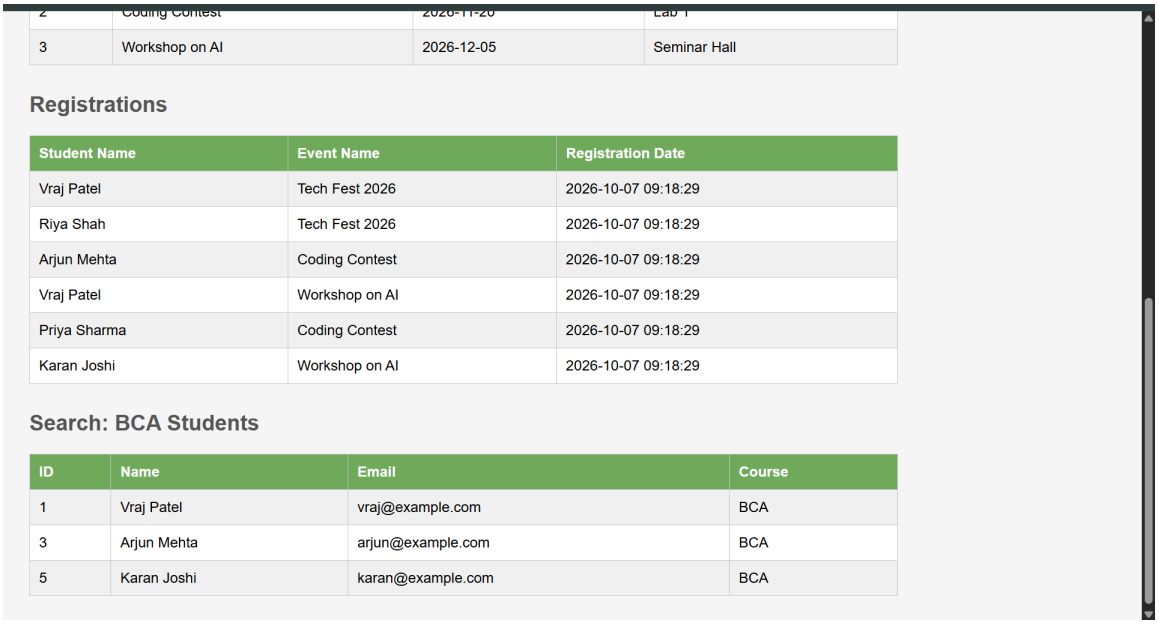
Database connected successfully!

All Students

ID	Name	Email	Course
1	Vraj Patel	vraj@example.com	BCA
2	Riya Shah	riya@example.com	MCA
3	Arjun Mehta	arjun@example.com	BCA
4	Priya Sharma	priya@example.com	MSc IT
5	Karan Joshi	karan@example.com	BCA

All Events

ID	Event Name	Date	Location
1	Tech Fest 2026	2026-11-15	Auditorium
2	Coding Contest	2026-11-20	Lab 1
3	Workshop on AI	2026-12-05	Seminar Hall



2	Coding Contest	2026-11-20	Lab 1
3	Workshop on AI	2026-12-05	Seminar Hall

Registrations

Student Name	Event Name	Registration Date
Vraj Patel	Tech Fest 2026	2026-10-07 09:18:29
Riya Shah	Tech Fest 2026	2026-10-07 09:18:29
Arjun Mehta	Coding Contest	2026-10-07 09:18:29
Vraj Patel	Workshop on AI	2026-10-07 09:18:29
Priya Sharma	Coding Contest	2026-10-07 09:18:29
Karan Joshi	Workshop on AI	2026-10-07 09:18:29

Search: BCA Students

ID	Name	Email	Course
1	Vraj Patel	vraj@example.com	BCA
3	Arjun Mehta	arjun@example.com	BCA
5	Karan Joshi	karan@example.com	BCA

A. GitHub Repository Link:

https://github.com/RahilBarchha/student-hub-.git

B. Challenges Faced:

I did not know much about MYSQL at the beginning.
I had difficulty understanding PHP syntax. And How to connect with MYSQL.
I was confused about how to connect my HTML pages with MYSQL.
I did not know how to run my project using the XAMPP server. Also How to Make Table in XAMMP Server also diificult for beginner .
I faced errors while connecting the HTML form with the PHP file. And MYSQL.

C.Learning Outcome:-

Learned to design a normalized MySQL database, connect PHP with MySQL, and use primary/foreign keys and prepared statements securely.